



CAR RENTAL MICROSERVICES SYSTEM APPLICATION MONITORING & CI/CD SERVER

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PROJECT OVERVIEW



- Microservice-based car rental system
- Independent services
- Dockerized environment
- CI/CD automation with Jenkins
- Monitoring with Datadog

TECHNOLOGIES USED



Node.js

Server-side JavaScript runtime for building scalable applications.



Docker

Containerization platform with Docker Compose orchestration.



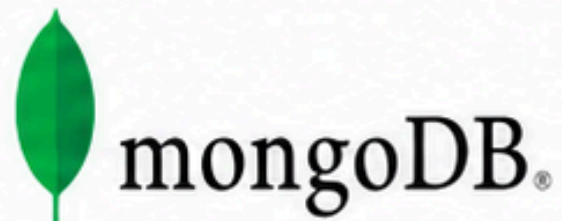
Express.js

Fast, minimalist web framework for Node.js REST APIs.



Jenkins

CI/CD automation server for continuous integration.



MongoDB

NoSQL database for flexible document storage.

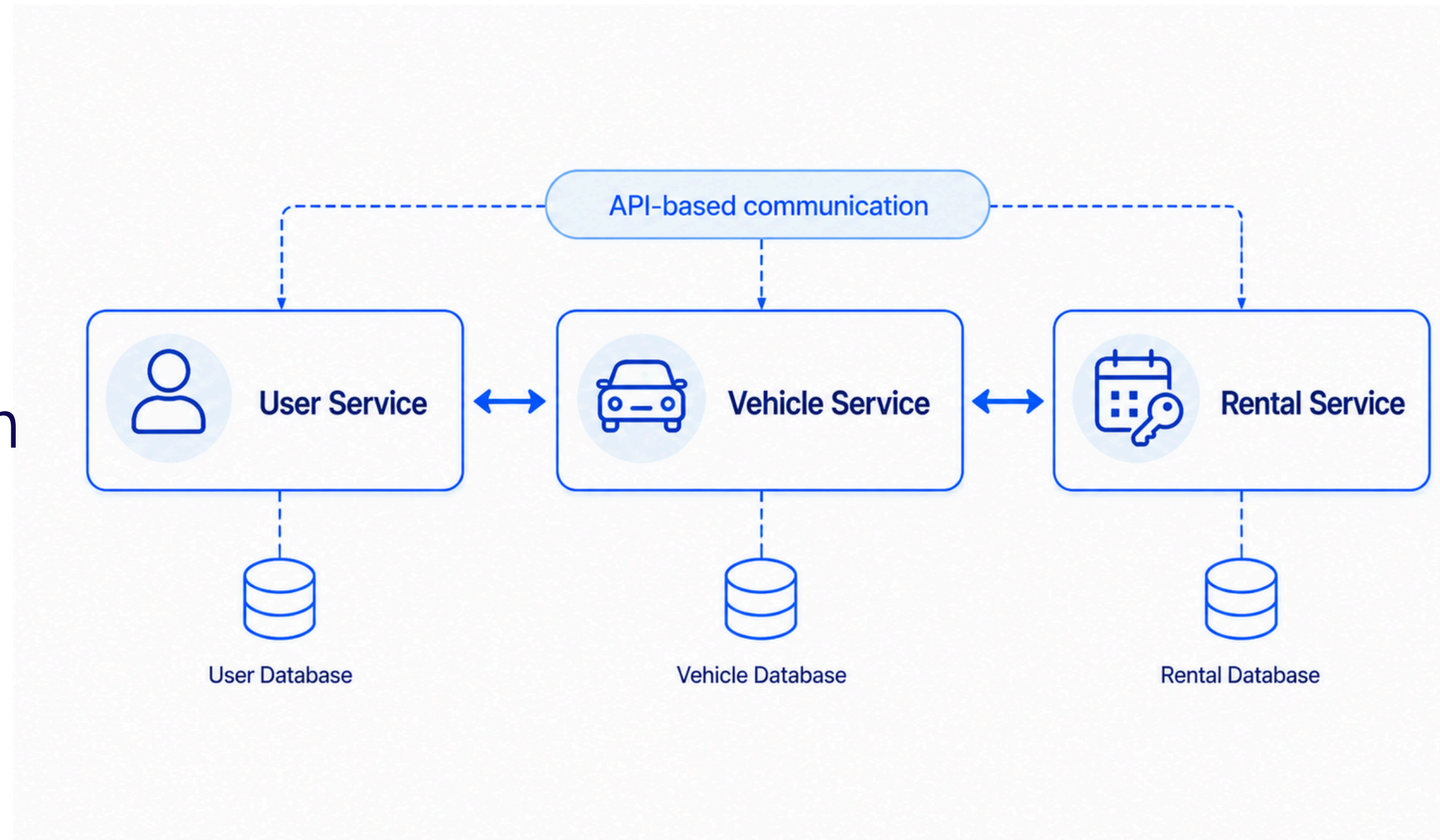


Datadog

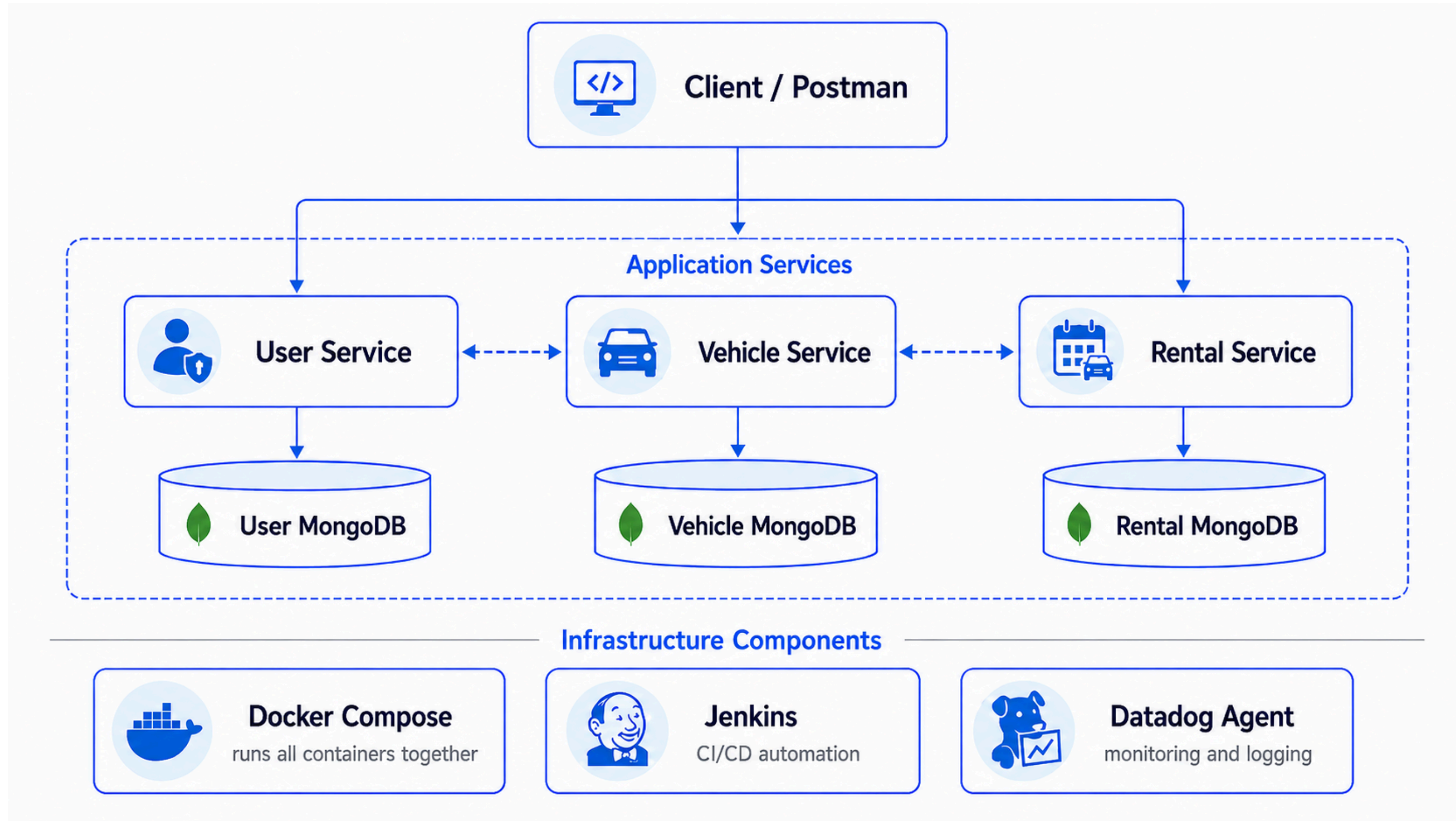
Application monitoring and observability platform.

MICROSERVICE ARCHITECTURE

- Independent services
- Separate responsibilities
- API-based communication
- Easier deployment
- Scalable system design



SYSTEM ARCHITECTURE

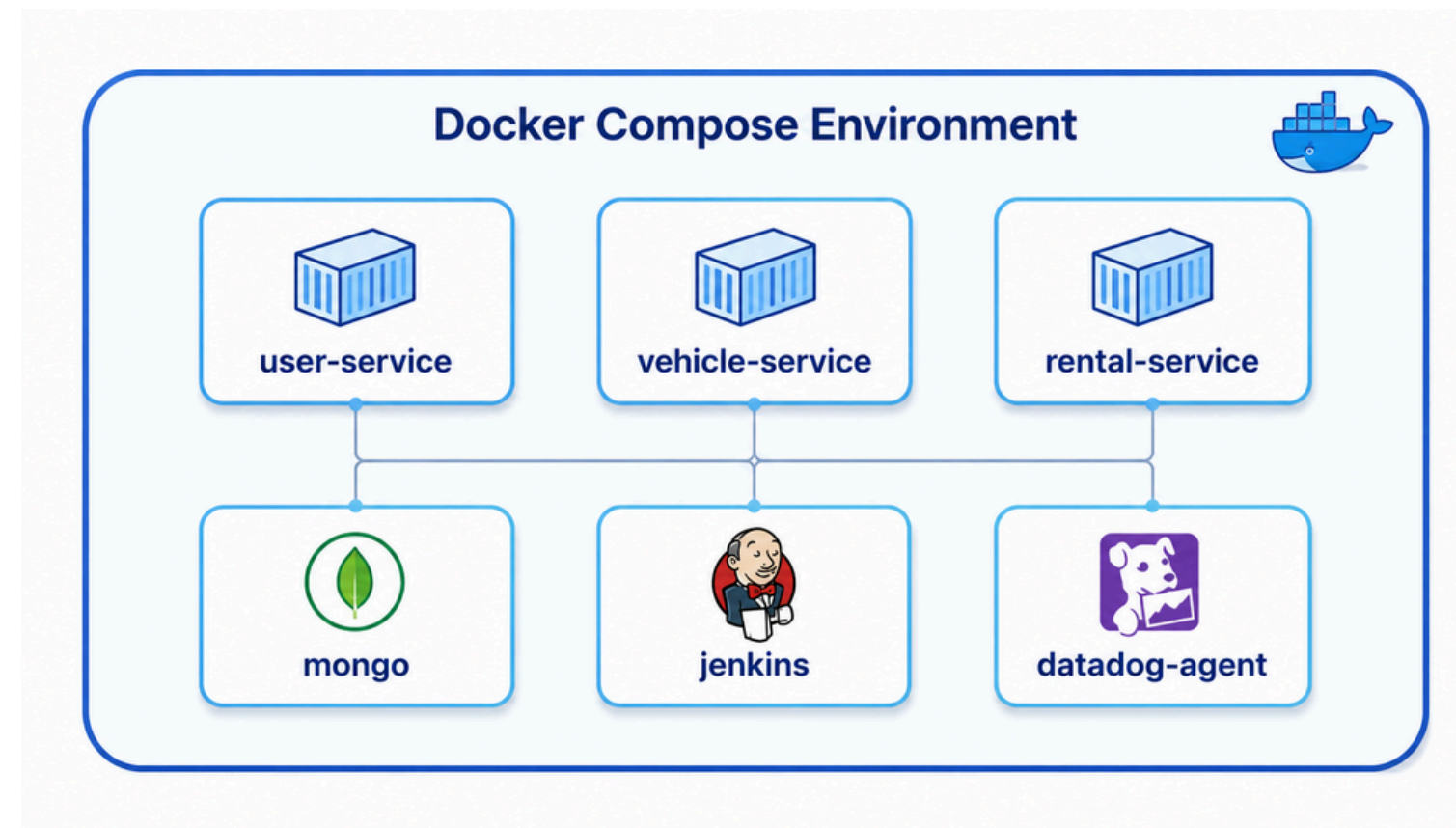


DOCKERIZED ENVIRONMENT

Docker is a containerization platform that allows applications to run in isolated environments.

In our project:

- Each service runs in a separate container
- MongoDB runs as a container
- Docker Compose manages all services
- Services communicate in the same network

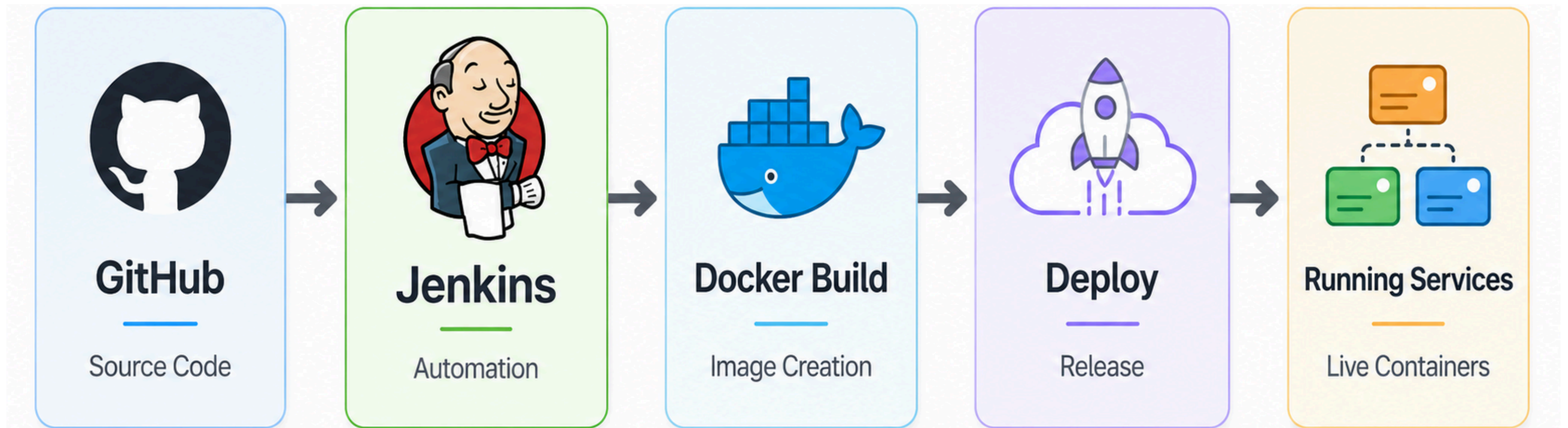


CI/CD WITH JENKINS

Jenkins is an automation server used to support CI/CD processes.

In our project, Jenkins automates:

- Source code checkout
- Docker image build
- Service deployment
- Pipeline execution

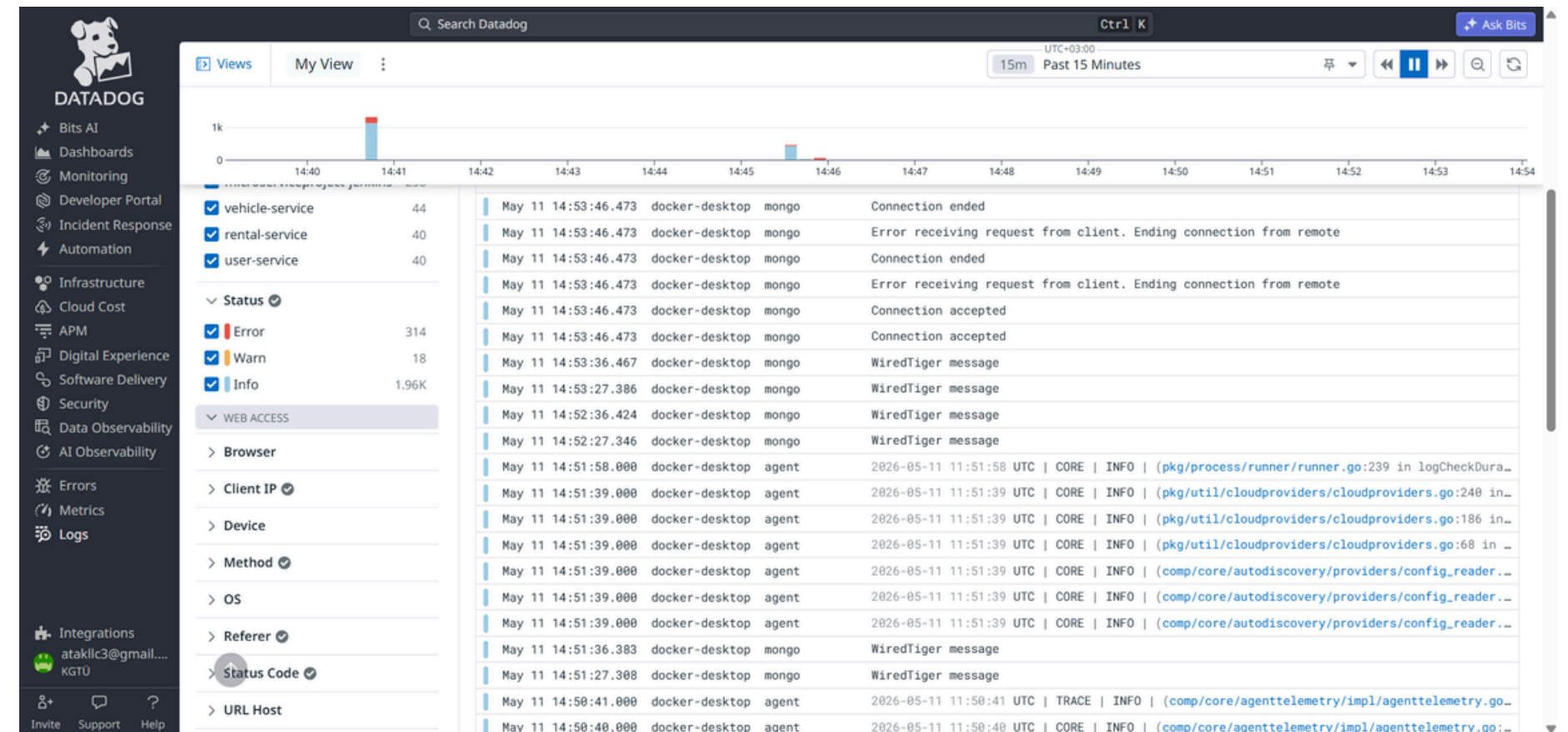


APPLICATION MONITORING WITH DATADOG

Datadog is a monitoring platform used to observe running applications and services.

It provides:

- Container status monitoring
- Service logs
- Runtime behavior observation
- Error tracking through logs
- Health check visibility



CONCLUSION & FUTURE WORK



Completed Features



Three Independent Microservices

User Service, Vehicle Service and Rental Service with separate responsibilities.



Dockerized Deployment

All services and infrastructure components run in isolated Docker containers.



Jenkins CI/CD Pipeline

Automated build, containerization and deployment pipeline implemented.



Datadog Monitoring Integration

Logs, metrics, traces and service health are monitored in real-time.



Secure API Communication

JWT authentication, role-based authorization and internal API key for service communication.



Future Improvements



API Gateway

Implement an API Gateway for unified entry point, routing, and security.



Frontend Interface

Develop a modern web interface for users to interact with the system.



Advanced Dashboards

Create custom monitoring dashboards and alerts in Datadog.



Automated Testing

Add automated unit, integration and end-to-end tests to the pipeline.



This project is a foundation for scalable, maintainable and observable systems.
We will continue to improve and extend it in the future.